

GAMBELL, AK, USA

WMO: 702040

Lat: 63.767N

Lon: 171.733W

Elev: 9

StdP: 101.22

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26703

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-23.8	-32.9	0.2	-20.0	-31.2	0.2	-21.2	22.4	-17.2	21.2	-17.7	7.6	40	1.122

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.6	12.5	10.7	11.9	10.4	11.1	9.8	11.2	12.0	10.4	11.3	9.8	10.7	5.5	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	8.0	11.6	9.6	7.4	10.9	8.9	7.1	10.4	32.0	11.9	29.9	11.3	28.3	10.6	14.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.6	17.9	16.1	DB	-27.0	15.4	3.4	1.7	-29.4	16.6	-31.3	17.6	-33.2	18.6	-35.7	19.9	
			WB	-27.1	11.5	3.3	1.8	-29.5	12.8	-31.4	13.8	-33.3	14.9	-35.7	16.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.4	-13.4	-11.5	-11.3	-7.3	-0.5	4.5	8.1	8.5	5.9	1.3	-3.3	-10.3
	DBStd	9.15	7.22	7.49	6.75	4.72	2.58	2.41	1.81	1.81	2.43	2.94	4.05	6.80
	HDD10.0	4531	726	601	659	520	324	165	63	53	123	271	398	628
	HDD18.3	7561	984	834	918	770	583	415	316	305	372	529	648	886
	CDD10.0	13	0	0	0	0	0	1	6	6	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.3	10.4	10.3	8.5	8.4	6.5	5.8	5.9	6.6	7.4	8.7	10.0	10.8
Precipitation	PrecAvg	581	42	39	29	26	30	37	60	82	72	62	59	45
	PrecMax	887	205	159	80	101	71	85	106	142	123	152	113	103
	PrecMin	347	1	5	0	4	10	8	18	16	29	13	10	1
	PrecStd	123	40	32	23	19	18	16	21	26	25	31	32	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	2.5	1.2	2.3	6.1	12.7	15.0	13.7	11.4	9.4	4.3	2.2
		MCWB	5.1	1.2	0.3	1.2	3.8	9.0	11.6	11.4	9.7	8.2	3.3	1.0
	2%	DB	2.2	1.3	0.4	1.2	3.8	10.6	12.7	12.4	10.9	8.0	3.0	1.0
		MCWB	1.0	0.3	-0.4	0.3	2.4	8.4	10.8	11.0	9.4	7.0	2.0	0.3
	5%	DB	-0.9	0.2	-0.5	0.2	2.8	8.9	11.9	12.0	9.9	7.1	2.5	-0.1
		MCWB	-1.5	-0.4	-1.3	-0.5	1.6	7.4	10.4	10.8	8.6	5.8	1.6	-0.6
	10%	DB	-2.6	-0.9	-1.6	-1.1	2.3	7.7	11.0	11.1	9.0	5.1	2.0	-1.3
		MCWB	-3.3	-1.4	-2.2	-1.6	1.2	6.3	9.7	10.0	7.8	3.7	1.2	-1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	1.6	0.7	1.4	4.2	10.1	12.3	12.1	10.3	8.5	3.6	1.0
		MCDB	8.0	2.0	1.0	1.7	6.2	11.9	14.0	12.9	10.9	9.1	4.2	1.4
	2%	WB	0.7	0.3	-0.3	0.2	2.8	8.6	11.3	11.4	9.6	7.5	2.5	0.2
		MCDB	1.6	0.9	0.3	0.8	3.7	10.0	12.3	12.0	10.3	8.1	3.0	0.7
	5%	WB	-1.3	-0.3	-1.0	-0.5	2.0	7.7	10.4	10.7	8.8	5.8	1.7	-0.6
		MCDB	-0.8	0.3	-0.5	0.1	2.7	8.8	11.5	11.3	9.5	7.1	2.3	-0.2
	10%	WB	-3.4	-1.3	-2.1	-1.6	1.3	6.7	9.6	10.0	8.0	4.0	0.8	-1.9
		MCDB	-2.6	-0.8	-1.6	-1.2	2.0	7.8	10.7	10.8	8.7	5.3	1.6	-1.5
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.6	4.8	3.9	2.9	3.4	3.0	2.6	2.3	2.2	2.6	3.4
		MCDBR	7.2	5.1	4.1	3.9	3.9	5.4	5.1	3.6	2.5	2.1	2.4	3.8
		MCWBR	6.5	5.0	3.9	3.7	3.1	3.6	3.2	2.6	2.0	2.0	2.7	3.8
	5% WB	MCDBR	7.2	5.0	4.0	3.8	3.6	4.8	4.2	3.0	2.2	2.0	2.5	3.6
		MCWBR	6.6	5.0	3.9	3.6	3.1	3.4	3.0	2.5	2.0	1.9	2.8	3.7
Clear-Sky Solar Irradiance	taub	0.185	0.268	0.282	0.348	0.384	0.368	0.379	0.390	0.337	0.304	0.403	0.385	
	taud	1.961	2.075	2.163	2.026	2.149	2.415	2.429	2.342	2.452	2.288	2.675	2.696	
	Ebn at Noon	427	639	808	813	819	851	828	775	754	602	311	189	
	Edh at Noon	28	65	91	132	129	102	98	98	72	54	28	18	
All-Sky Solar Radiation	RadAvg	0.17	0.85	2.53	4.12	4.47	5.28	4.25	3.13	2.03	0.84	0.19	0.06	
	RadStd	0.02	0.13	0.25	0.46	0.48	0.38	0.38	0.34	0.18	0.10	0.02	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page